

Bathymetry Dataflow

M. Schumacher 01/2024

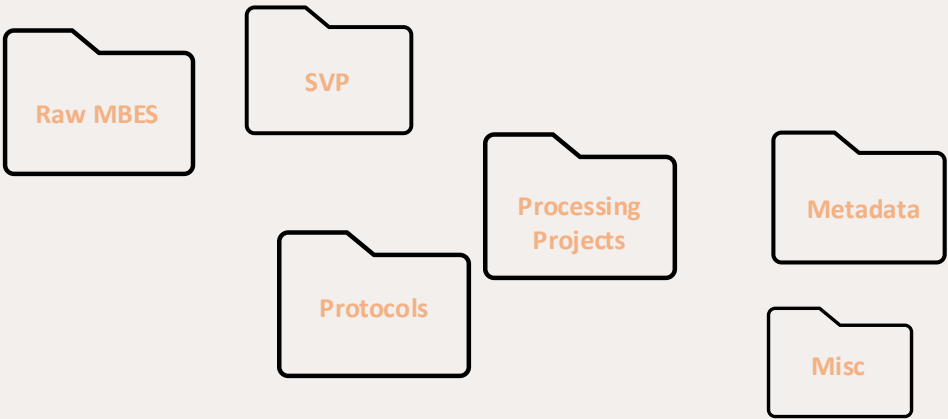


On Board

Acquisition



Cruise Leader

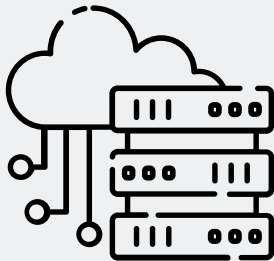
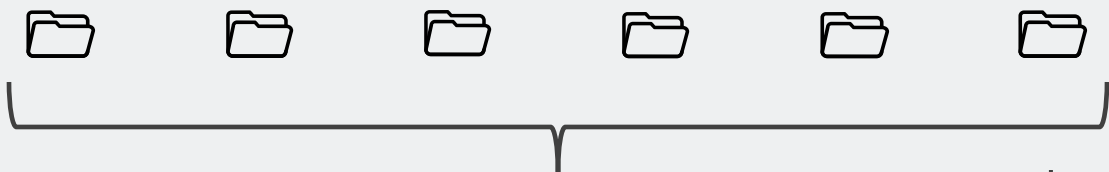


Back Home

Management



**GEOMAR and Data
MGMT Bathymetrists**



- Sorting*
- Quality check
- Push to internal DB*

* Automisable if uniform on- board structure

• Raw data via DAM

Service



**GEOMAR
Bathymetrists**

- Processing
- Data compilation
- Data products:
 - .gfs/.esf
 - .xyz
 - .tif/.nc



Publication

- Products and processed raw data (.gfs)



GEOMAR Bathymetrists

Legend



On board folder structure



Database



GEOMAR Grid



Person responsible

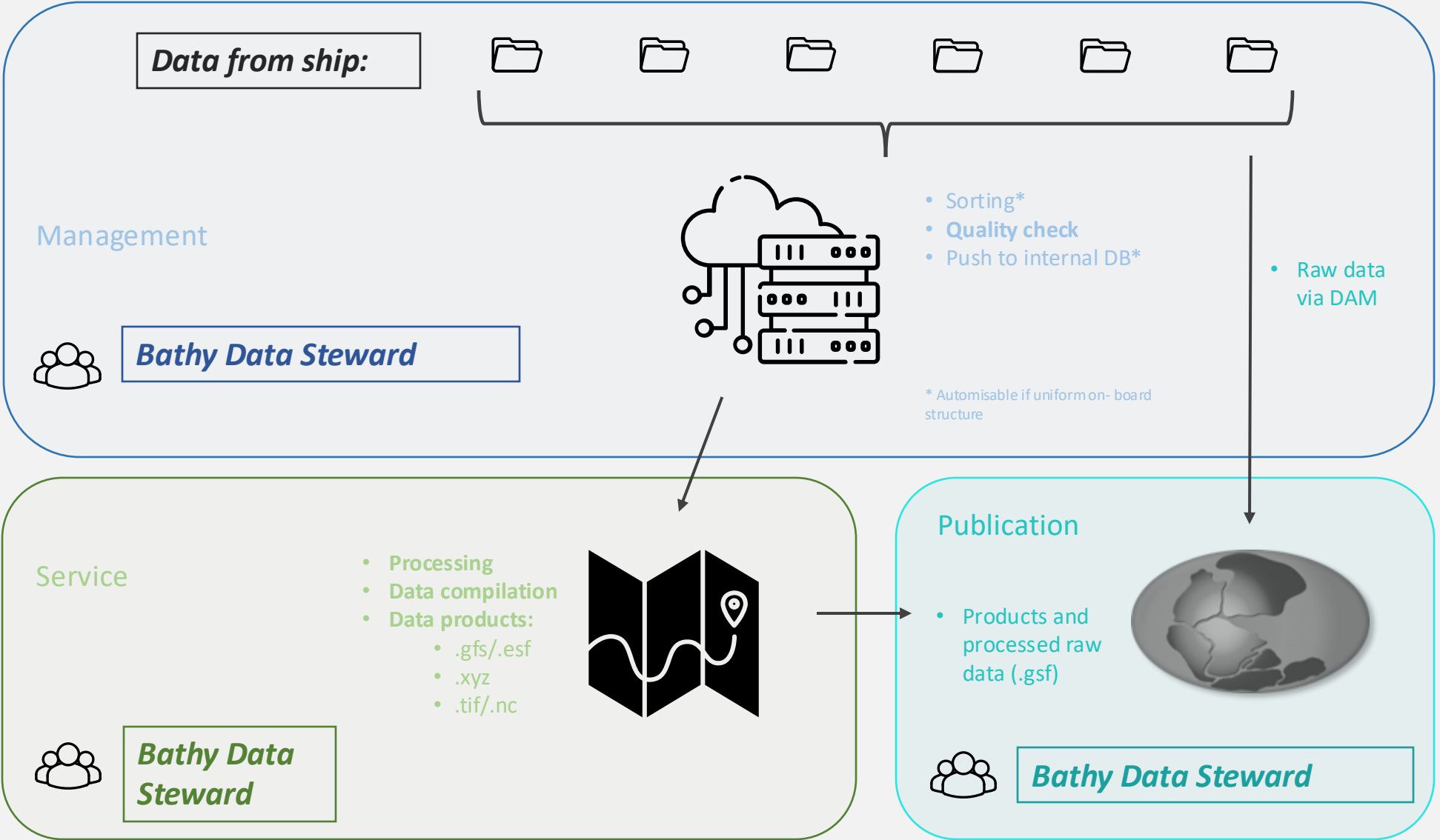
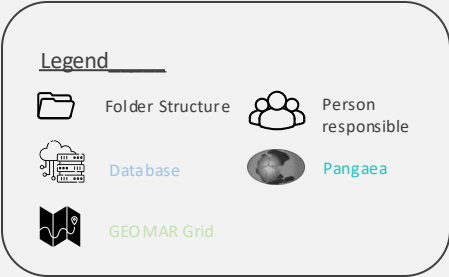


Pangaea

Bathymetry Dataflow

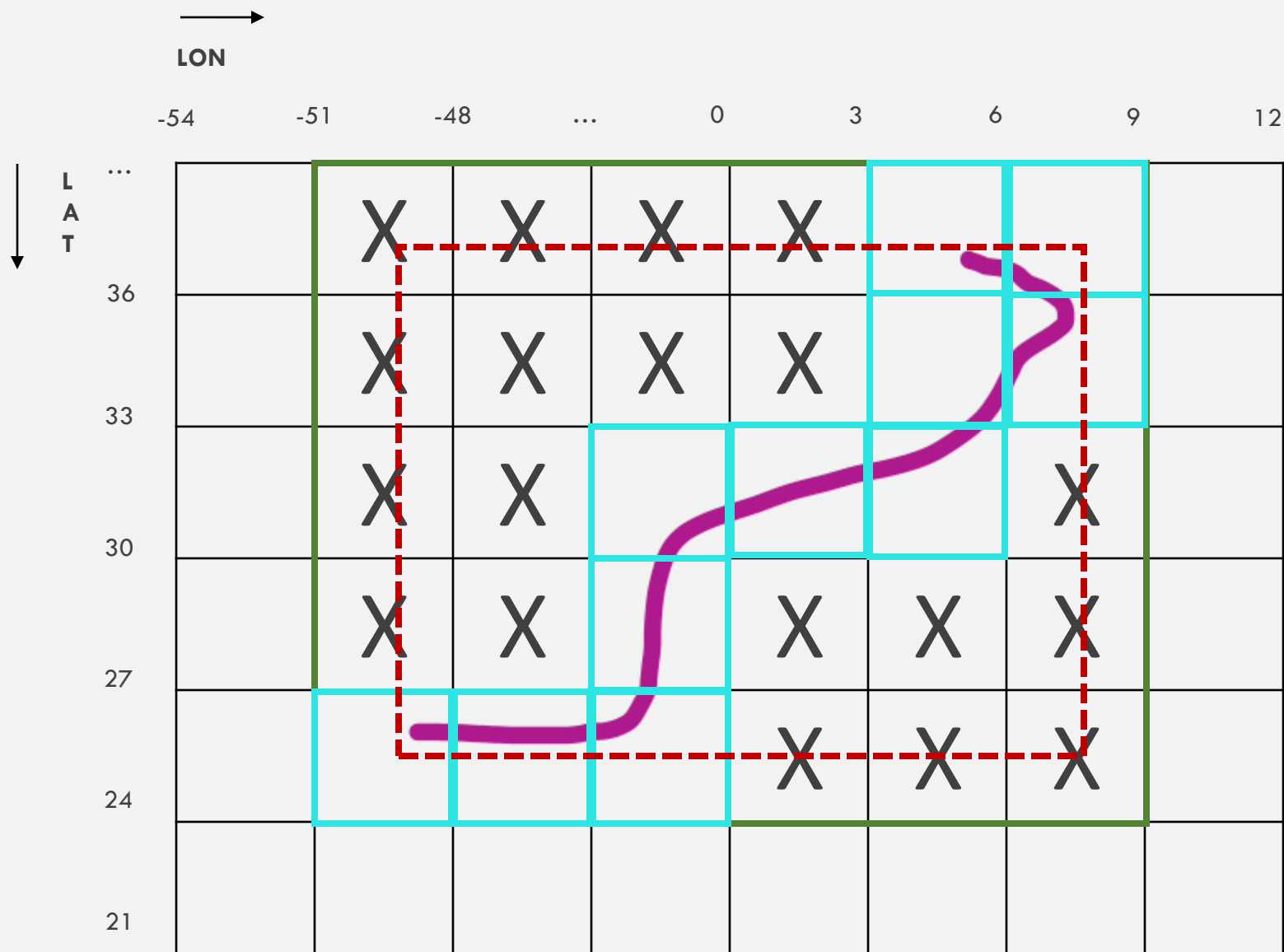
M. Schumacher 01/2024

Bathymetry Data Steward (M. Schumacher)



>		00_MARIASMERIAN
>		00_METEOR
▼		00_SONNE
		DIPL_Genehmigungen_SONNE_Stand-2020-10-30.zip
>		Other_Institutes
▼		SONNE_GEOMAR
>		SO96_1
>		SO96_2
>		SO97
>		SO97_2
>		SO201
>		SO233
>		SO234
>		SO237
>		SO241
>		SO243_1
>		SO249
>		SO249_2
>		SO252
>		SO254
>		SO255
>		SO258_1
>		SO264
>		SO265
>		SO267
▼		SO273
▼		_metadata
>		_misc
>		_protocols
>		_svp
▼		raw
		SO273_fileinfo.txt
▼		SO273_products
>		_grd
>		_gsf
▼		_xyz
		SO273_xyz_accepted_UTM_32S.txt
>		SO273_Qimera
>		SO277
>		SO279
>		SO283

Folder Structure smb://filer



Cruise track/Bathymetry swath



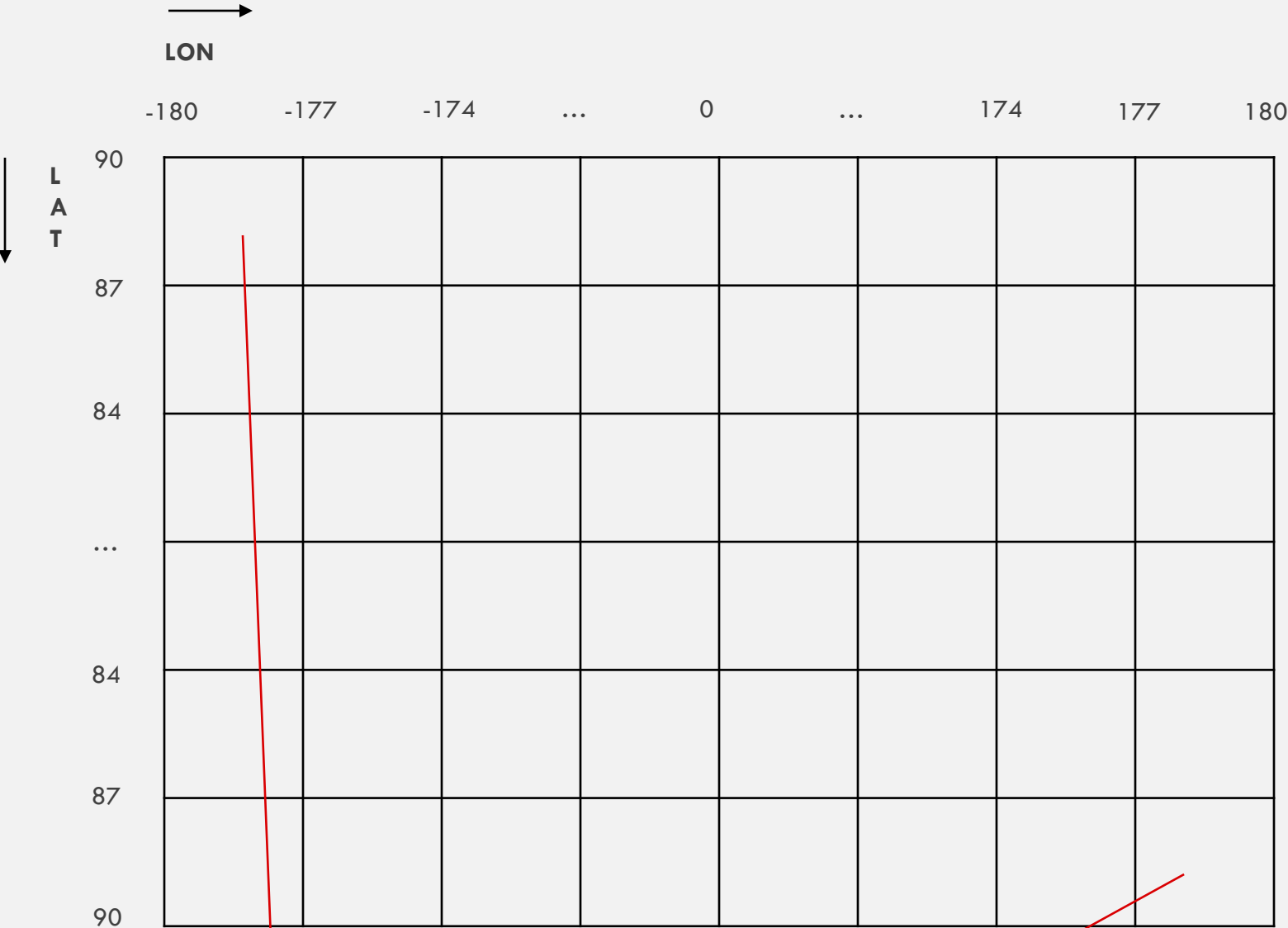
Cruise Region



Cruise Tile Region



Cruise Region chunks



GlobalTiles_3deg

Tile_177E-180E_87N-90N.xyz

Tile_174E-177E_87N-90N.xyz

Tile_171E-174E_87N-90N.xyz

Tile_168E-171E_87N-90N.xyz

Tile_165E-168E_87N-90N.xyz

Tile_162E-165E_87N-90N.xyz

Tile_159E-162E_87N-90N.xyz

Tile_156E-159E_87N-90N.xyz

Tile_153E-156E_87N-90N.xyz

Tile_150E-153E_87N-90N.xyz

Tile_147E-150E_87N-90N.xyz

Tile_144E-147E_87N-90N.xyz

...

Tile_141W-138W_90S-87S.xyz

Tile_144W-141W_90S-87S.xyz

Tile_147W-144W_90S-87S.xyz

Tile_150W-147W_90S-87S.xyz

Tile_153W-150W_90S-87S.xyz

Tile_156W-153W_90S-87S.xyz

Tile_159W-156W_90S-87S.xyz

Tile_162W-159W_90S-87S.xyz

Tile_165W-162W_90S-87S.xyz

Tile_168W-165W_90S-87S.xyz

Tile_171W-168W_90S-87S.xyz

Tile_174W-171W_90S-87S.xyz

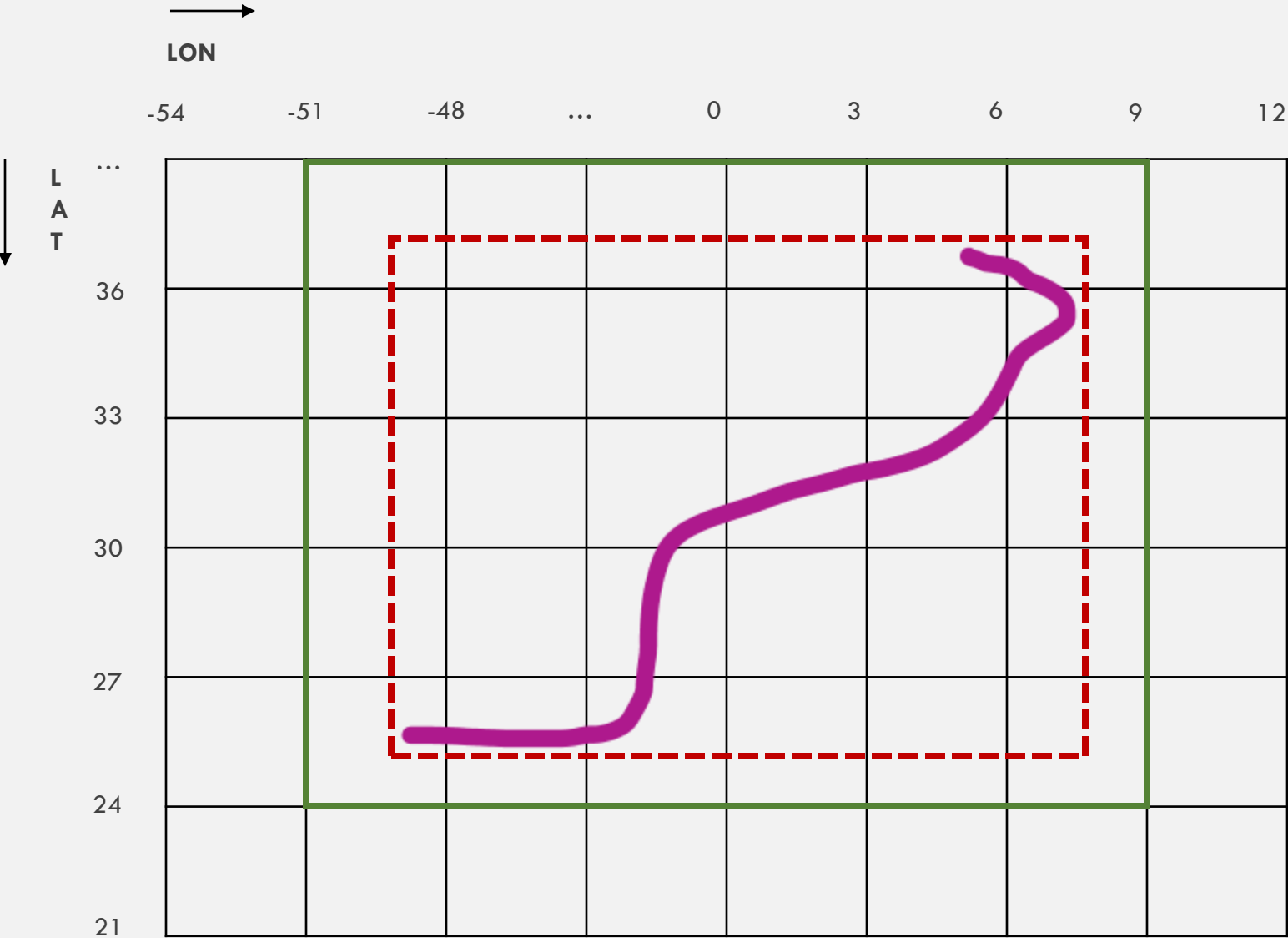
Tile_177W-174W_90S-87S.xyz

Tile_180W-177W_90S-87S.xyz

- -R177/180/87/90
- -R174/177/87/90
- -R171/174/87/90
- -R174/-171/-90/-87
- -R177/-174/-90/-87
- -R180/-177/-90/-87

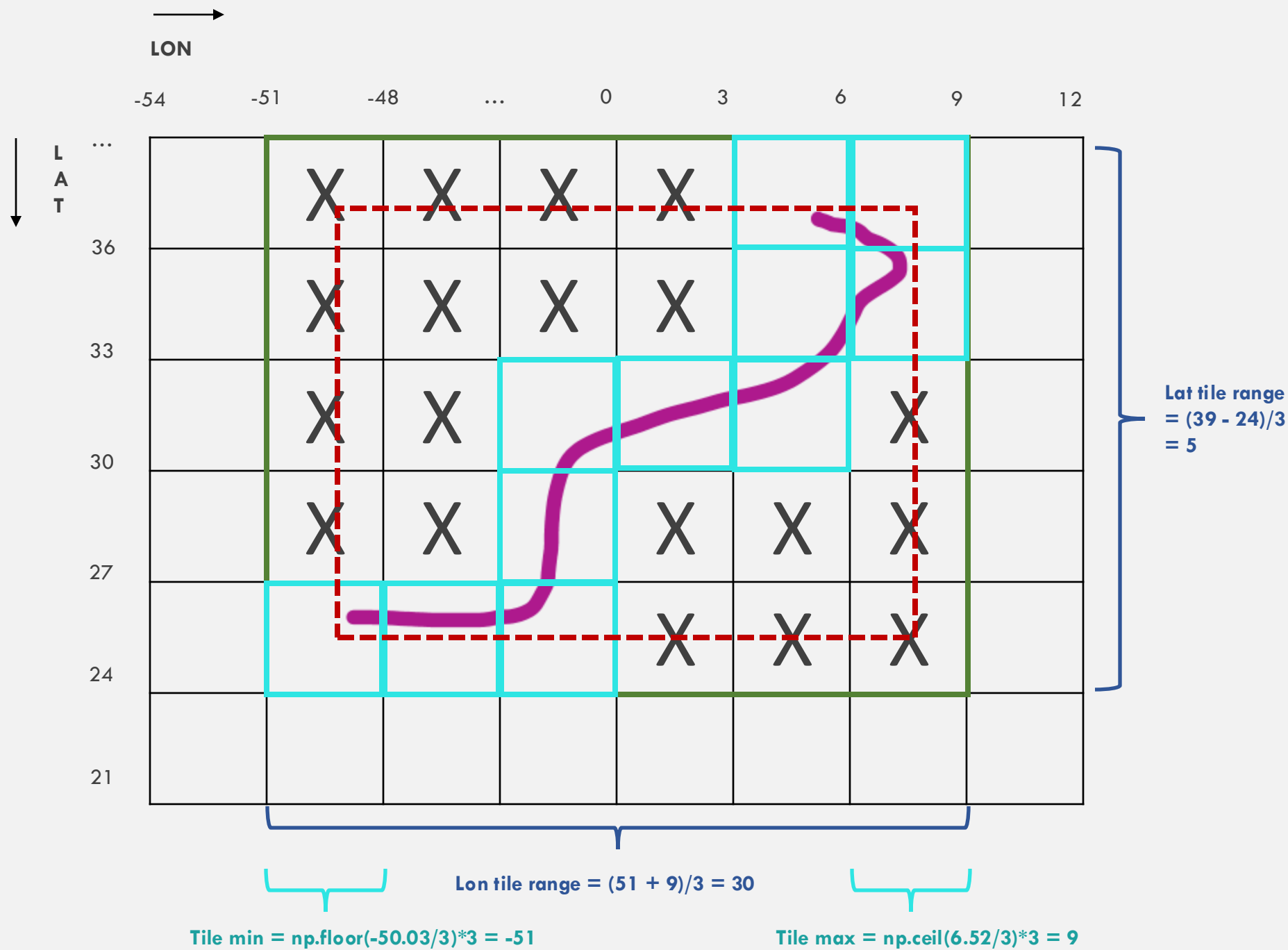
Empty Tile Files

List of Global Tile Regions



With gmt info, get regions from each .xyz file and write to region.reg (red dashed line)
-> Fit region into area of 3x3° grid (solid green line)

Break up .xyz into chunks based on its tile regions: 02_Tile_XYZ.py



- ,Widen' region to 3x3° grid: Calculate min/max tile range
- Calculate number of tiles in tile range
- Get tile chunks: Calculate number of tiles in each direction (lon, lat) within tile range

Tilenum_min/max = (np.floor/ceil(min/max_xyz/tile_size)) * tile_size with tile_size = 3 (degree)

Tile range = abs(int(tilenum_min - tilenum_max)/tile_size)

Lon/Lat_chunk = lon/lat_tilenum + (rx * tile_size) with r = tilenum at index

Cont.: Break up .xyz into chunks based on its tile regions: 02_Tile_XYZ.py

Lon; Lat; Depth

-50.03847135; 26.78541190; -4067.81

-50.03861337; 26.78541356; -4061.18

-50.03869476; 26.78541356; -4047.02

-50.03899363; 26.78541853; -4056.14

-50.03921012; 26.78542151; -4056.84

-50.03848420; 26.78481762; -4066.85

-50.03862918; 26.78482193; -4060.48

-49.03877812; 26.78482656; -4053.16

-49.03905261; 26.78483385; -4061.13

-49.03738296; 26.78539633; -4065.73

...

4.52224805; 34.94635685; -4740.81

4.52208626; 34.94688422; -4732.94

4.52198213; 34.94722409; -4720.85

4.51902935; 34.94602691; -4686.45

5.51895389; 34.94629490; -4683.01

5.51892621; 35.94639627; -4690.28

5.51890974; 35.94646120; -4703.69

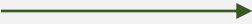
6.51882307; 35.94676629; -4697.53

6.51877991; 35.94692132; -4700.10

6.51872488; 35.94711710; -4699.56

GMT info: -R-50/7/26/37

Wider tile region



-R-51 /9/24/39

Region chunks



-R-51 /-48/24/27

-R-48/-45/24/27

-R-45/-42/24/27

...

-R-51 /-48/27/30

-R-51 /-48/30/33

-R-51 /-48/33/36

...

-R6/9/36/39

Distribute .xyz chunks to respective tile file: 02_GMT_Distribute_XYZ_Tiles.sh

Lon; Lat; Depth

-50.03847135; 26.78541190; -4067.81

-50.03861337; 26.78541356; -4061.18

-50.03869476; 26.78541356; -4047.02

-50.03899363; 26.78541853; -4056.14

-50.03921012; 26.78542151; -4056.84

-50.03848420; 26.78481762; -4066.85

-50.03862918; 26.78482193; -4060.48

-49.03877812; 26.78482656; -4053.16

-49.03905261; 26.78483385; -4061.13

-49.03738296; 26.78539633; -4065.73

...

4.52224805; 34.94635685; -4740.81

4.52208626; 34.94688422; -4732.94

4.52198213; 34.94722409; -4720.85

4.51902935; 34.94602691; -4686.45

5.51895389; 34.94629490; -4683.01

5.51892621; 35.94639627; -4690.28

5.51890974; 35.94646120; -4703.69

6.51882307; 35.94676629; -4697.53

6.51877991; 35.94692132; -4700.10

6.51872488; 35.94711710; -4699.56



Tile_045W-042W_27N-30N.xyz

Tile_048W-045W_27N-30N.xvz

Tile_051W-048W_27N-30N.xyz

Tile_054W-051W_27N-30N.xyz

Tile_057W-054W_27N-30N.xyz

Tile_060W-057W_27N-30N.xyz

Tile_063W-060W_27N-30N.xyz

Tile_066W-063W_27N-30N.xyz

Tile_069W-066W_27N-30N.xvz

...

Tile_021E-024E_33N-36N.xyz

Tile_018E-021E_33N-36N.xyz

Tile_015E-018E_33N-36N.xyz

Tile_012E-015E_33N-36N.xyz

Tile_009E-012E_33N-36N.xvz

Tile_006E-009E_33N-36N.xyz

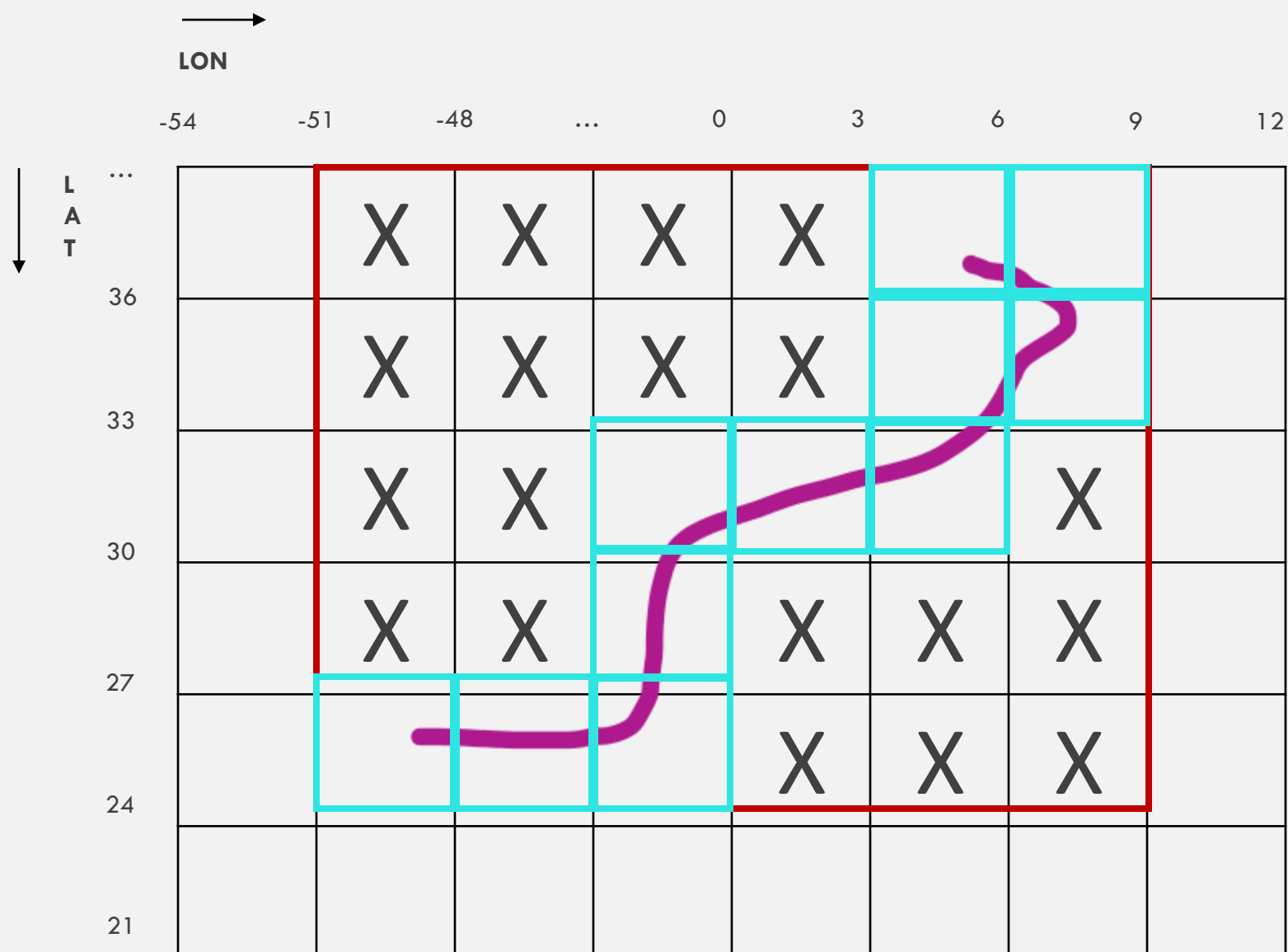
Tile_003E-006E_33N-36N.xyz

Tile_000E-003E_33N-36N.xyz

Tile_003W-000W_33N-36N.xyz

Distribute xyz to correct Tile Files

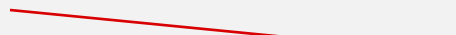
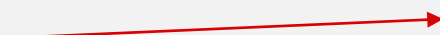
Get Regions from .xyz: 00b_GMT_Extract_Regions.sh



Distribute .xyz chunks to respective tile file: 02_GMT_Distribute_XYZ_Tiles.sh

Lon; Lat; Depth

-50.038471135; 26.78541190; -4067.81
-50.03861337; 26.78541356; -4061.18
-50.03869476; 26.78541356; -4047.02
-50.03899363; 26.78541853; -4056.14
-50.03921012; 26.78542151; -4056.84
-50.03848420; 26.78481762; -4066.85
-50.03862918; 26.78482193; -4060.48
-49.03877812; 26.78482656; -4053.16
-49.03905261; 26.78483385; -4061.13
-49.03738296; 26.78539633; -4065.73
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4.52198213; 34.94722409; -4720.85
4.51902935; 34.94602691; -4686.45
5.51895389; 34.94629490; -4683.01
5.51892621; 35.94639627; -4690.28
5.51890974; 35.94646120; -4703.69
6.51882307; 35.94676629; -4697.53
6.51877991; 35.94692132; -4700.10
6.51872488; 35.94711710; -4699.56



- Tile_045W-042W_27N-30N.xyz
- Tile_048W-045W_27N-30N.xyz
- Tile_051W-048W_27N-30N.xyz
- Tile_054W-051W_27N-30N.xyz
- Tile_057W-054W_27N-30N.xyz
- Tile_060W-057W_27N-30N.xyz
- Tile_063W-060W_27N-30N.xyz
- Tile_066W-063W_27N-30N.xyz
- Tile_069W-066W_27N-30N.xyz

...

Tile_021E-024E_33N-36N.xyz
Tile_018E-021E_33N-36N.xyz
Tile_015E-018E_33N-36N.xyz
Tile_012E-015E_33N-36N.xyz
Tile_009E-012E_33N-36N.xyz
Tile_006E-009E_33N-36N.xyz
Tile_003E-006E_33N-36N.xyz
Tile_000E-003E_33N-36N.xyz
Tile_003W-000W_33N-36N.xyz

Distribute xyz to correct Tile Files